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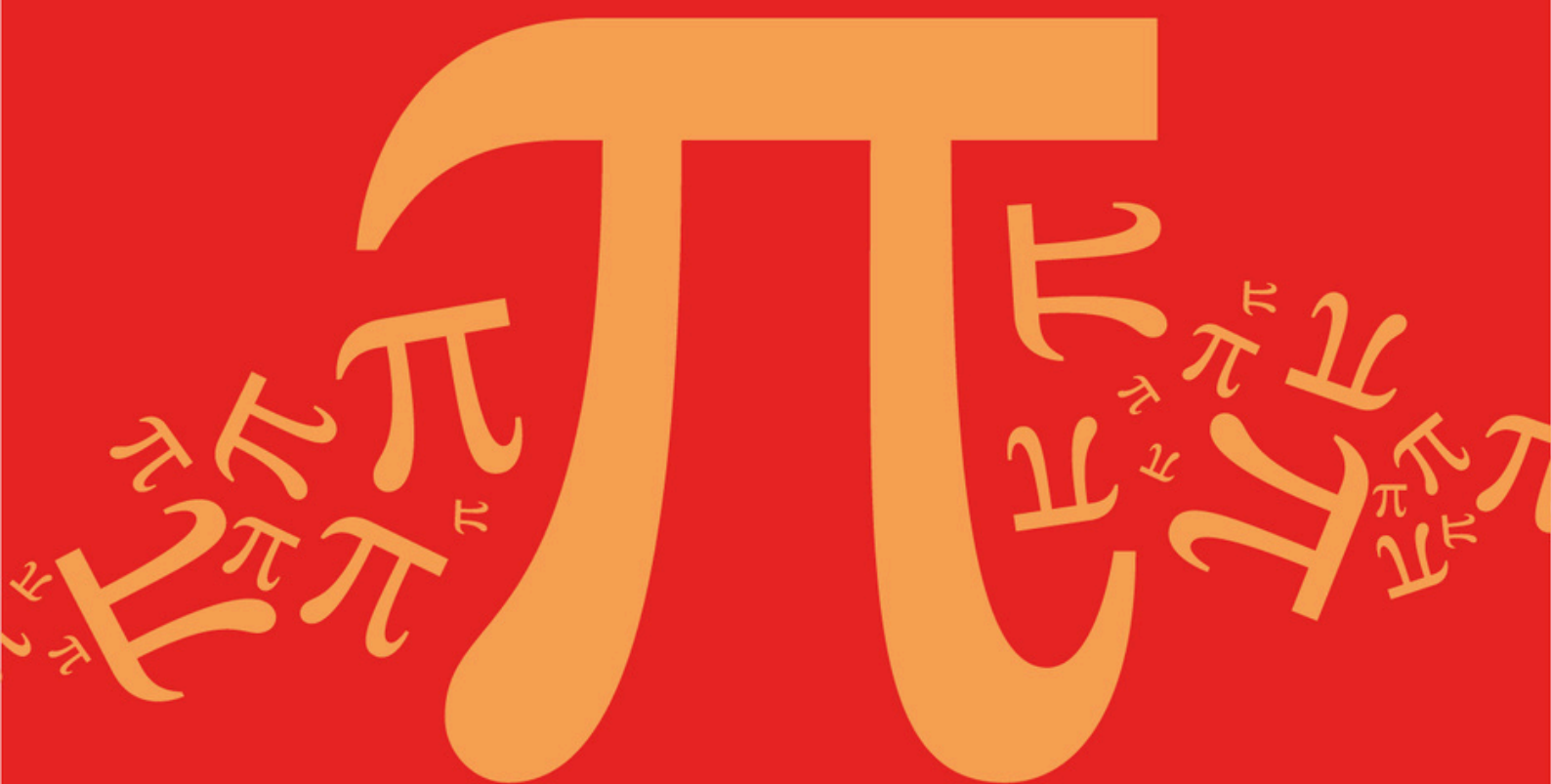
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AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C3: Matrix Transformations (2D)

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

C: Matrices

Sub topic

C3: Matrix Transformations (2D)

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**To be used by all students and teachers for
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Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

Find two invariant points under the transformation given by $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$

(Total 2 marks)

Q2.

A reflection is represented by the matrix $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

State the equation of the line of invariant points.

Circle your answer.

$x = 0$

$y = 0$

$y = x$

$y = -x$

(Total 1 mark)

Q3.

$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & k \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

(a) Find the value of k for which matrix $\mathbf{A}$ is singular.

(1)

(b) Describe the transformation represented by matrix $\mathbf{B}$.

(1)

(c) (i) Given that $\mathbf{A}$ and $\mathbf{B}$ are both non-singular, verify that $\mathbf{A}^{-1}\mathbf{B}^{-1} = (\mathbf{BA})^{-1}$.

(4)

(ii) Prove the result $\mathbf{M}^{-1}\mathbf{N}^{-1} = (\mathbf{NM})^{-1}$ for all non-singular matrices $\mathbf{M}$ and $\mathbf{N}$.

(4)

(Total 10 marks)

Q4.

(a) The matrix $\mathbf{X}$ is defined by $\begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$.

(i) Given that $\mathbf{X}^2 = \begin{bmatrix} m & 2 \\ 3 & 6 \end{bmatrix}$, find the value of m .

(1)

(ii) Show that $\mathbf{X}^3 - 7\mathbf{X} = n\mathbf{I}$, where n is an integer and $\mathbf{I}$ is the 2×2 identity matrix.

(4)

(b) It is given that $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

(i) Describe the geometrical transformation represented by $\mathbf{A}$.

(1)

(ii) The matrix $\mathbf{B}$ represents an anticlockwise rotation through 45° about the origin.

Show that $\mathbf{B} = k \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$, where k is a surd.

(2)

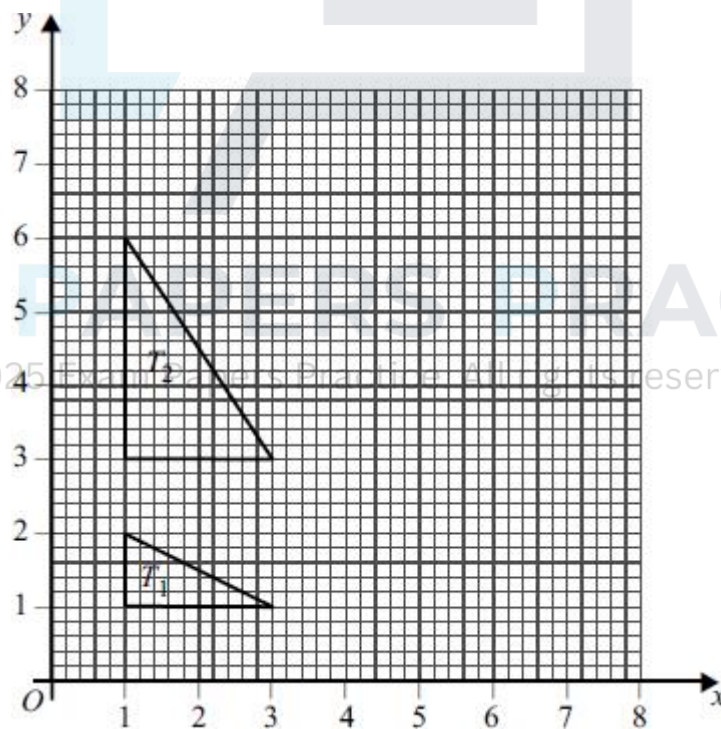
(iii) Find the image of the point $P(-1, 2)$ under an anticlockwise rotation through 45° about the origin, followed by the transformation represented by $\mathbf{A}$.

(4)

(Total 12 marks)

Q5.

The diagram shows two triangles, T_1 and T_2 .



(a) Find the matrix which represents the stretch that maps triangle T_1 onto triangle T_2 .

(2)

(b) The triangle T_2 is reflected in the line $y = \sqrt{3}x$ to give a third triangle, T_3 . Find, using surd forms where appropriate:

(i) the matrix which represents the reflection that maps triangle T_2 onto triangle T_3 ;

(2)

(ii) the matrix which represents the combined transformation that maps triangle T_1

onto triangle T_3 .

(2)

(Total 6 marks)

Q6.

The plane transformation T is defined by

$$T : \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ -3 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- (a) A shape has an area of 3 square units. Find the area of the shape after being transformed by T .

(2)

- (b) (i) Find the equations of all the invariant lines of T .

(5)

- (ii) State the equation of the line of invariant points of T .

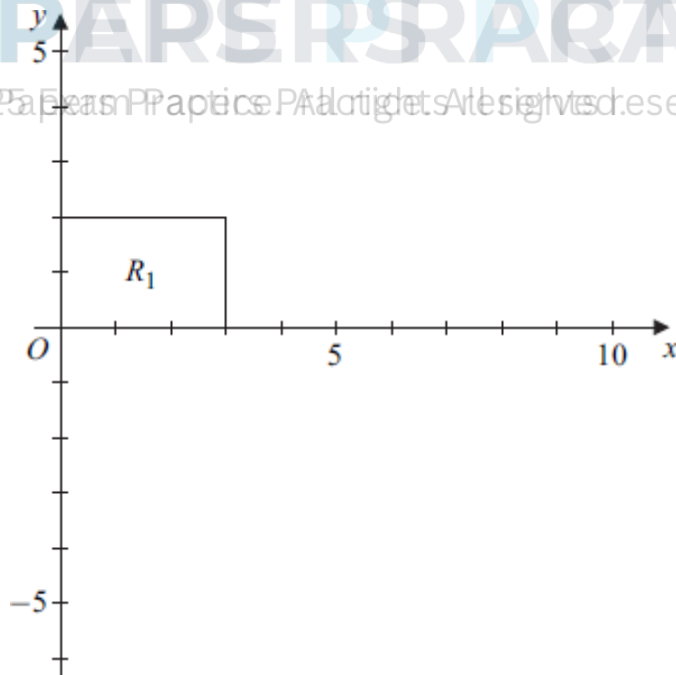
(1)

(Total 8 marks)

Q7.

The diagram below shows a rectangle R_1 which has vertices $(0, 0)$, $(3, 0)$, $(3, 2)$ and $(0, 2)$.

- (a) On the diagram, draw:



- (i) the image R_2 of R_1 under a rotation through 90° clockwise about the origin;

(1)

- (ii) the image R_3 of R_2 under the transformation which has matrix

$$\begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$$

(3)

(b) Find the matrix of:

(i) the rotation which maps R_1 onto R_2 ;

(1)

(ii) the combined transformation which maps R_1 onto R_3 .

(3)

(Total 8 marks)

Q8.

The plane transformation T is a rotation through θ radians anticlockwise about O , and maps points (x, y) onto image points (X, Y) such that

$$\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} c & -s \\ s & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

where $c = \cos \theta$ and $s = \sin \theta$.

(a) Write down the inverse of the matrix $\begin{bmatrix} c & -s \\ s & c \end{bmatrix}$ and hence show that

$$x = cX + sY \quad \text{and} \quad y = -sX + cY$$

(3)

(b) The curve C has equation $x^2 - 6xy - 7y^2 = 8$.

The image of C under T is the curve C' with equation $pX^2 + qXY + rY^2 = 8$.

(i) Use the results of part (a) to show that

$$q = 6s^2 + 16sc - 6c^2$$

and express p and r similarly in terms of c and s .

(4)

(ii) Given that θ is an acute angle, find the values of c and s for which $q = 0$ and hence in this case express the equation of C' in the form

$$\frac{X^2}{a^2} - \frac{Y^2}{b^2} = 1$$

(8)

(iii) Hence explain why C is a hyperbola.

(1)

(Total 16 marks)

Q9.

- (a) Using surd forms, find the matrix of a rotation about the origin through 135° anticlockwise.

(2)

(b) The matrix $\mathbf{M}$ is defined by $\mathbf{M} = \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix}$.

- (i) Given that $\mathbf{M}$ represents an enlargement followed by a rotation, find the scale factor of the enlargement and the angle of the rotation.

(3)

- (ii) The matrix $\mathbf{M}^2$ also represents an enlargement followed by a rotation. State the scale factor of the enlargement and the angle of the rotation.

(2)

- (iii) Show that $\mathbf{M}^4 = k\mathbf{I}$, where k is an integer and $\mathbf{I}$ is the 2×2 identity matrix.

(2)

- (iv) Deduce that $\mathbf{M}^{2012} = -2^n\mathbf{I}$ for some positive integer n .

(2)

(Total 11 marks)

Q10.

- (a) Write down the 2×2 matrix corresponding to each of the following transformations:

- (i) a rotation about the origin through 90° clockwise;

(1)

- (ii) a rotation about the origin through 180° .

(1)

- (b) The matrices $\mathbf{A}$ and $\mathbf{B}$ are defined by

$$\mathbf{A} = \begin{bmatrix} 2 & 4 \\ -1 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -2 & 1 \\ -4 & 3 \end{bmatrix}$$

- (i) Calculate the matrix $\mathbf{AB}$.

(2)

- (ii) Show that $(\mathbf{A} + \mathbf{B})^2 = k\mathbf{I}$, where $\mathbf{I}$ is the identity matrix, for some integer k .

(3)

- (c) Describe the single geometrical transformation, or combination of two geometrical transformations, represented by each of the following matrices:

- (i) $\mathbf{A} + \mathbf{B}$;

(2)

- (ii) $(\mathbf{A} + \mathbf{B})^2$;

(2)

(iii) $(\mathbf{A} + \mathbf{B})^4$.

(2)

(Total 13 marks)

Q11.

The plane transformation T is represented by the matrix $\mathbf{M} = \begin{bmatrix} -3 & 8 \\ -1 & 3 \end{bmatrix}$.

- (a) The quadrilateral $ABCD$ has image $A'B'C'D'$ under T .

Evaluate $\det \mathbf{M}$ and describe the geometrical significance of both its sign and its magnitude in relation to $ABCD$ and $A'B'C'D'$.

(3)

- (b) The line $y = px$ is a line of invariant points of T , and the line $y = qx$ is an invariant line of T .

Show that $p = \frac{1}{2}$ and determine the value of q .

(5)

- (c) Find the 2×2 matrix $\mathbf{R}$ which represents a reflection in the line $y = \frac{1}{2}x$.

(2)

(Total 10 marks)

Q12.

The matrix $\mathbf{A}$ is defined by

$$\mathbf{A} = \begin{bmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{bmatrix}$$

- (a) (i) Calculate the matrix $\mathbf{A}^2$

(2)

- (ii) Show that $\mathbf{A}^3 = k\mathbf{I}$, where k is an integer and $\mathbf{I}$ is the 2×2 identity matrix.

(2)

- (b) Describe the single geometrical transformation, or combination of two geometrical transformations, corresponding to each of the matrices:

- (i) $\mathbf{A}^3$;

(2)

- (ii) $\mathbf{A}$.

(3)

(Total 9 marks)

Q13.

The plane transformation T is the composition of a reflection in the line $y = x \tan \alpha$ followed by an anticlockwise rotation about O through an angle β .

Determine the matrix which represents T , and hence describe T as a single transformation.

(Total 6 marks)

Q14.

(a) The transformation U of three-dimensional space is represented by the matrix

$$\begin{bmatrix} 1 & 4 & -3 \\ 2 & -1 & 0 \\ 1 & 1 & -1 \end{bmatrix}$$

(i) Write down a vector equation for the line L with cartesian equation

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{6}$$

(2)

(ii) Find a vector equation for the image of L under U , and deduce that it is a line through the origin.

(4)

(b) The plane transformation V is represented by the matrix $\begin{bmatrix} 1 & 4 \\ 2 & -1 \end{bmatrix}$.

L_1 is the line with equation $y = \frac{1}{2}x + k$ and L_2 is the image of L_1 under V .

(i) Find, in the form $y = mx + c$, the cartesian equation for L_2

(4)

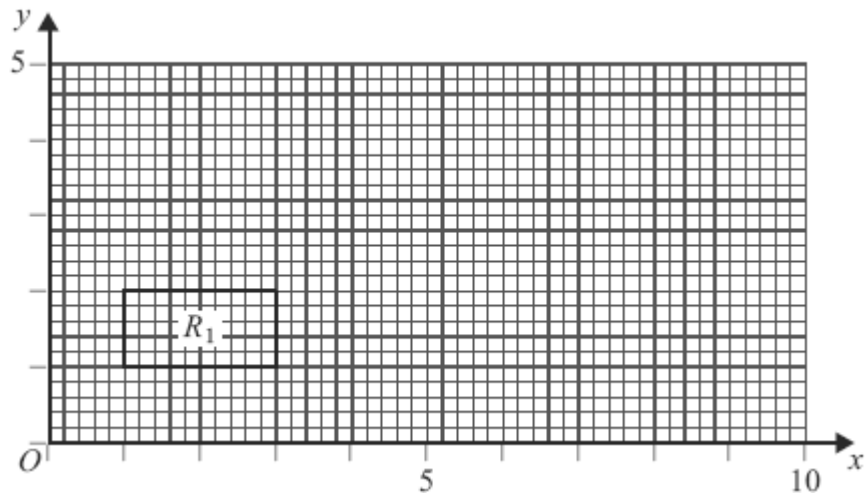
(ii) Deduce that L_2 is parallel to L_1 and find, in terms of k , the distance between these two lines.

(3)

(Total 13 marks)

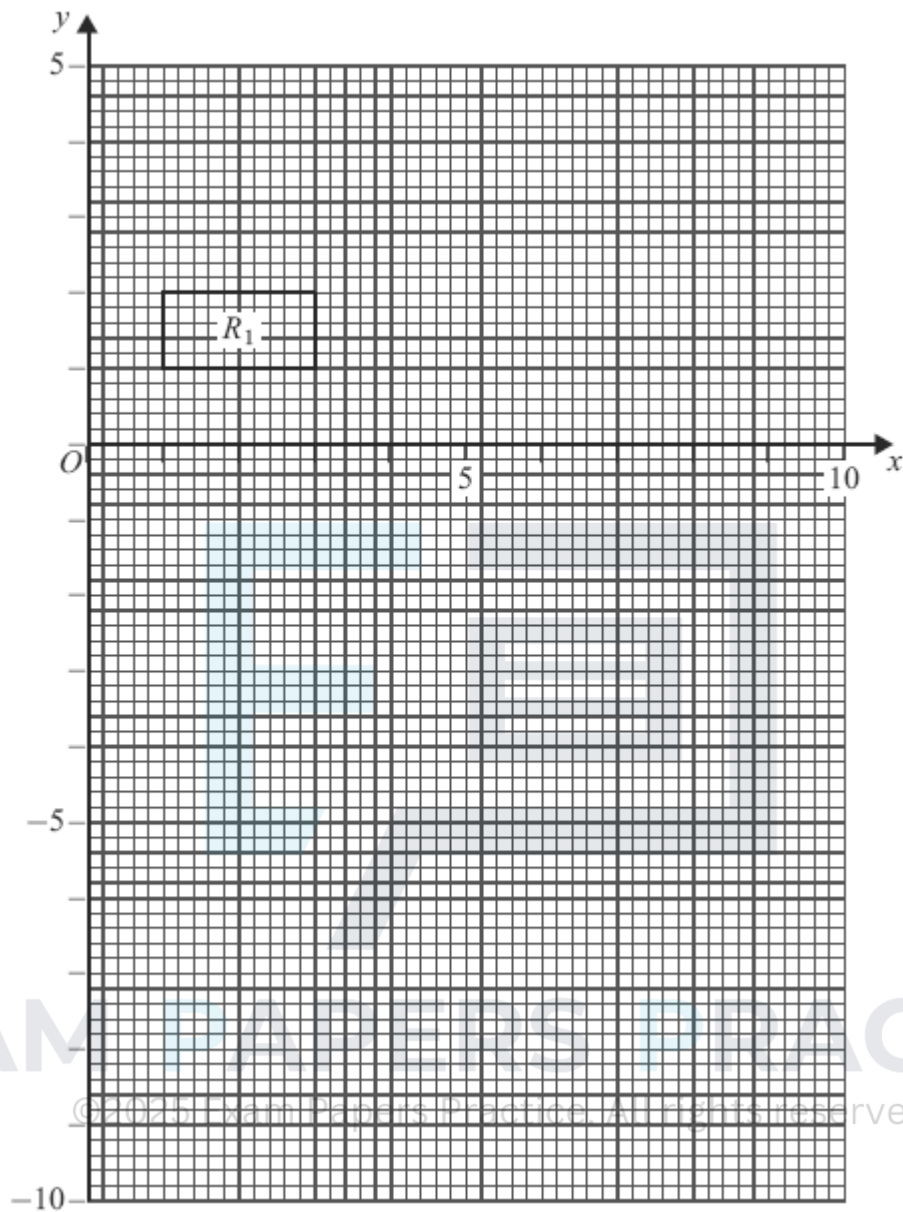
Q15.

The diagram shows a rectangle R_1 .



- (a) The rectangle R_1 is mapped onto a second rectangle, R_2 , by a transformation with matrix $\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$.
- (i) Calculate the coordinates of the vertices of the rectangle R_2 .
- (ii) On the figure below, draw the rectangle R_2 .

(2)



(1)

- (b) The rectangle R_2 is rotated through 90° clockwise about the origin to give a third rectangle, R_3 .

- (i) On the figure above, draw the rectangle R_3 .

(2)

- (ii) Write down the matrix of the rotation which maps R_2 onto R_3 .

(1)

- (c) Find the matrix of the transformation which maps R_1 onto R_3 .

(2)

(Total 8 marks)

Q16.

The 2×2 matrix $\mathbf{M}$ represents the plane transformation T . Write down the value of $\det \mathbf{M}$ in each of the following cases:

- (a) T is a rotation;
- (b) T is a reflection;
- (c) T is an enlargement with scale factor 3.

(Total 3 marks)

Q17.

The plane transformations T_A and T_B are represented by the matrices $\mathbf{A}$ and $\mathbf{B}$

respectively, where $\mathbf{A} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 2 & 5 \\ 5 & 13 \end{bmatrix}$.

- (a) Find the equation of the line which is the image of $y = 2x + 1$ under T_A . (3)
- (b) The rectangle $PQRS$, with area 4.5 cm^2 , is mapped onto the parallelogram $P'Q'R'S'$ under T_B . Determine the area of $P'Q'R'S'$. (2)
- (c) The transformation T_C is the composition
' T_B followed by T_A '

Find the matrix which represents T_C , give a full geometrical description of T_C .

(5)

(Total 10 marks)

Q18.

The matrices $\mathbf{A}$ and $\mathbf{B}$ are defined by

$$\mathbf{A} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 1 & 1 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 1 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Describe fully the geometrical transformation represented by each of the following matrices:

- (a) $\mathbf{A}$; (2)
- (b) $\mathbf{B}$; (2)
- (c) $\mathbf{A}^2$; (2)
- (d) $\mathbf{B}^2$; (2)
- (e) $\mathbf{AB}$.

(3)
(Total 11 marks)

Q19.

The matrices **A** and **B** are defined by

$$\mathbf{A} = \begin{bmatrix} k & k \\ k & -k \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -k & k \\ k & k \end{bmatrix}$$

where k is a constant.

(a) Find, in terms of k :

(i) $\mathbf{A} + \mathbf{B}$;

(1)

(ii) $\mathbf{A}^2$.

(2)

(b) Show that $(\mathbf{A} + \mathbf{B})^2 = \mathbf{A}^2 + \mathbf{B}^2$.

(4)

(c) It is now given that $k = 1$.

(i) Describe the geometrical transformation represented by the matrix $\mathbf{A}^2$.

(2)

(ii) The matrix **A** represents a combination of an enlargement and a reflection. Find the scale factor of the enlargement and the equation of the mirror line of the reflection.

(3)

(Total 12 marks)

Q20.

The plane transformation **T** has matrix $\mathbf{A} = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$, and maps points (x, y) onto image points (X, Y) such that

$$\begin{bmatrix} X \\ Y \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \end{bmatrix}$$

(a) (i) Find $\mathbf{A}^{-1}$.

(2)

(ii) Hence express each of x and y in terms of X and Y .

(2)

(b) Give a full geometrical description of **T**.

(5)

- (c) Any plane curve with equation of the form $\frac{x^2}{p} + \frac{y^2}{q} = 1$, where p and q are distinct positive constants, is an ellipse.

(i) Show that the curve E with equation $6x^2 + y^2 = 3$ is an ellipse.

(1)

(ii) Deduce that the image of the curve E under T has equation

$$2X^2 + 4XY + 5Y^2 = 15$$

(2)

(iii) Explain why the curve with equation $2x^2 + 4xy + 5y^2 = 15$ is an ellipse.

(1)

(Total 13 marks)

Q21.

(a) Using surd forms where appropriate, find the matrix which represents:

(i) a rotation about the origin through 30° anticlockwise;

(2)

(ii) a reflection in the line $y = \frac{1}{\sqrt{3}}x$.

(2)

(b) The matrix $\mathbf{A}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & \sqrt{3} \\ \sqrt{3} & -1 \end{bmatrix}$$

represents a combination of an enlargement and a reflection. Find the scale factor of the enlargement and the equation of the mirror line of the reflection.

(2)

(c) The transformation represented by $\mathbf{A}$ is followed by the transformation represented by $\mathbf{B}$, where

$$\mathbf{B} = \begin{bmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{bmatrix}$$

Find the matrix of the combined transformation and give a full geometrical description of this combined transformation.

(5)

(Total 11 marks)

Q22.

The matrix $\mathbf{M}$ is defined by

$$\mathbf{M} = \begin{bmatrix} \sqrt{3} & 3 \\ 3 & -\sqrt{3} \end{bmatrix}$$

- (a) (i) Show that

$$\mathbf{M}^2 = p\mathbf{I}$$

where p is an integer and $\mathbf{I}$ is the 2×2 identity matrix.

(3)

- (ii) Show that the matrix $\mathbf{M}$ can be written in the form

$$q \begin{bmatrix} \cos 60^\circ & \sin 60^\circ \\ \sin 60^\circ & -\cos 60^\circ \end{bmatrix}$$

where q is a real number. Give the value of q in surd form.

(3)

- (b) The matrix $\mathbf{M}$ represents a combination of an enlargement and a reflection.

Find:

- (i) the scale factor of the enlargement;

(1)

- (ii) the equation of the mirror line of the reflection.

(1)

- (c) Describe fully the geometrical transformation represented by $\mathbf{M}^4$.

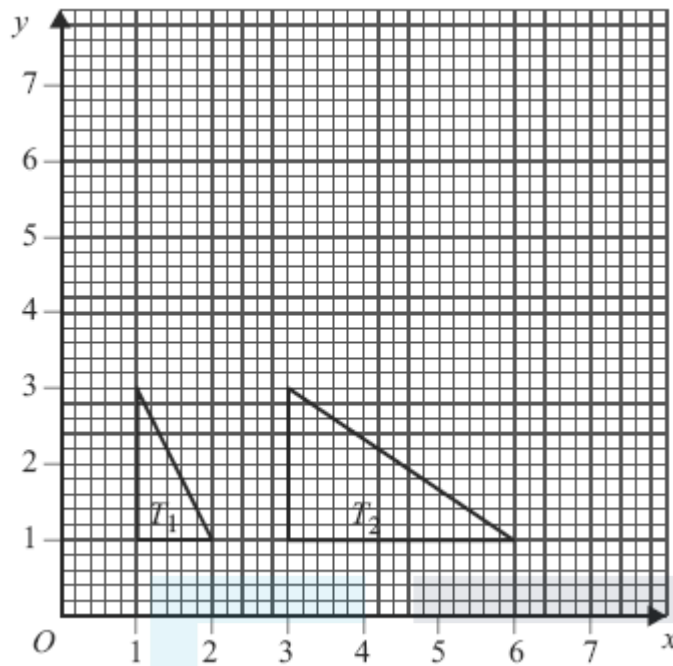
(2)

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(Total 10 marks)

Q23.

The diagram shows two triangles, T_1 and T_2 .



- (a) Find the matrix of the stretch which maps T_1 to T_2 . (2)
- (b) The triangle T_2 is reflected in the line $y = x$ to give a third triangle, T_3 .
On the figure above, draw the triangle T_3 . (2)
- (c) Find the matrix of the transformation which maps T_1 to T_3 . (3)

(Total 7 marks)

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Q24.

The matrices **A** and **B** are given by

$$\mathbf{A} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix}$$

- (a) Calculate:
- (i) $\mathbf{A} + \mathbf{B}$; (2)
- (ii) $\mathbf{BA}$. (3)
- (b) Describe fully the geometrical transformation represented by each of the following matrices:
- (i) $\mathbf{A}$; (2)

(ii) **B**;

(2)

(iii) **BA**.

(2)

(Total 11 marks)

Q25.

The matrices **A** and **B** are given by

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 3 & 8 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

The matrix **M** = **A** – 2**B**.

(a) Show that $\mathbf{M} = n \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$, where n is a positive integer.

(2)

(b) The matrix **M** represents a combination of an enlargement of scale factor p and a reflection in a line L . State the value of p and write down the equation of L .

(2)

(c) Show that

$$\mathbf{M}^2 = q\mathbf{I}$$

where q is an integer and **I** is the 2×2 identity matrix.

(2)

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Q26.

(a) The transformation **T** is defined by the matrix **A**, where

$$\mathbf{A} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

(i) Describe the transformation **T** geometrically.

(2)

(ii) Calculate the matrix product **A**².

(2)

(iii) Explain briefly why the transformation **T** followed by **T** is the identity transformation.

(1)

(b) The matrix **B** is defined by

$$\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

- (i) Calculate $\mathbf{B}^2 - \mathbf{A}^2$.

(3)

- (ii) Calculate $(\mathbf{B} + \mathbf{A})(\mathbf{B} - \mathbf{A})$.

(3)

(Total 11 marks)

Q27.

The transformation T maps (x, y) to (x', y') where

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- (a) Describe the difference between an *invariant line* and a *line of invariant points* of T .

(1)

- (b) Evaluate the determinant of the matrix $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ and describe the geometrical significance of the result in relation to T .

(2)

- (c) Show that T has a line of invariant points, and find a cartesian equation for this line.

(2)

- (d) (i) Find the image of the point $(x, -x + c)$ under T .

(2)

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- (ii) Hence show that all lines of the form $y = -x + c$, where c is an arbitrary constant, are invariant lines of T .

(2)

- (e) Describe the transformation T geometrically.

(3)

(Total 12 marks)

Q28.

The matrix $\mathbf{M}$ is defined by

$$\mathbf{M} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

- (a) Find the matrix:

- (i) $\mathbf{M}^2$;

(3)

- (ii) M^4 . (1)
- (b) Describe fully the geometrical transformation represented by M . (2)
- (c) Find the matrix M^{2006} . (3)
- (Total 9 marks)

Q29.

- (a) The transformation T_1 is defined by the matrix

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Describe this transformation geometrically.

- (b) The transformation T_2 is an anticlockwise rotation about the origin through an angle of 60° .

Find the matrix of the transformation T_2 . Use surds in your answer where appropriate.

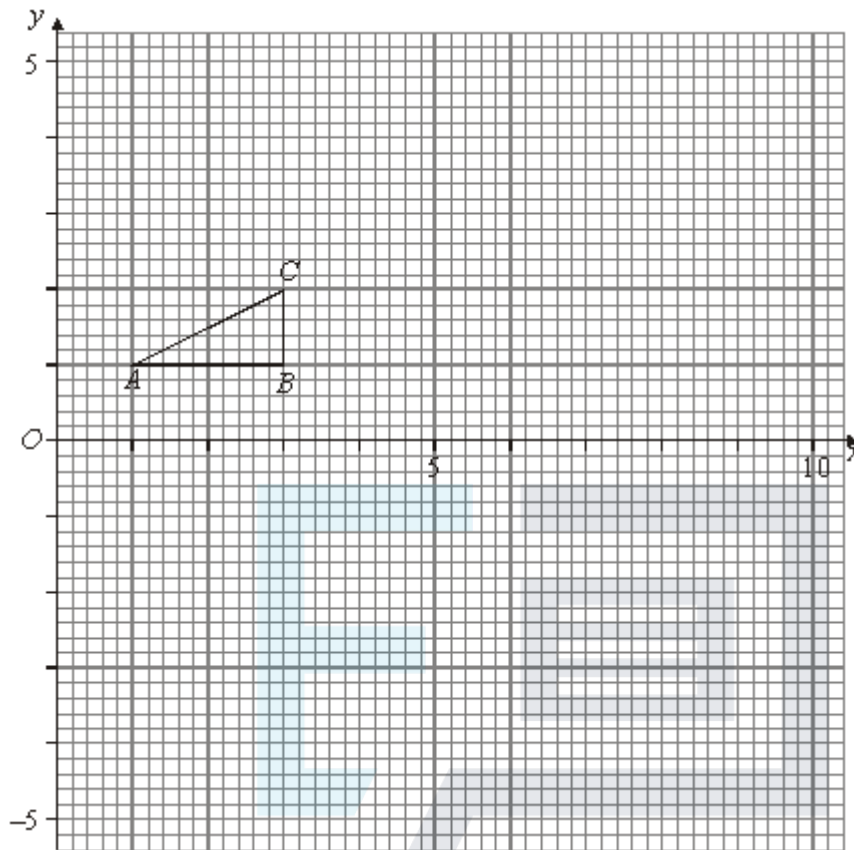
- (c) Find the matrix of the transformation obtained by carrying out T_1 followed by T_2 .

(3)
(Total 8 marks)

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Q30.

The diagram below shows a triangle with vertices $A(1, 1)$, $B(3, 1)$ and $C(3, 2)$.



- (a) The triangle DEF is obtained by applying to triangle ABC the transformation T represented by the matrix

$$\begin{bmatrix} 2 & 2 \\ -2 & 2 \end{bmatrix}$$

- (i) Calculate the coordinates of D , E and F . (4)
- (ii) Draw the triangle DEF on the diagram above. (2)
- (b) Given that T is a combination of an enlargement and a rotation, find the exact value of:
- (i) the scale factor of the enlargement; (2)
- (ii) the magnitude of the angle of the rotation. (2)

(Total 10 marks)

Q31.

The transformation T is a stretch in a fixed direction, and maps the point (x, y) to the image point (x', y') , where

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 6.4 & -7.2 \\ -7.2 & 10.6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- (a) Determine the scale factor of the stretch.

(2)

- (b) By considering the invariant points of T, or otherwise, find an equation for the line of invariant points of T, giving your answer in the form $y = mx$.

(5)

(Total 7 marks)



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